MTH 137() Dr. James Ashe
Math Lab Project **Name:**

Quadratic Function Project

1. Find a parabola in a work of art or architecture. Take or find a picture that gives a good frontal view of the parabola. (Here is a website you may want to take a look at: http://compfight.com/search/parabola/1-2-1-1.)

2. Draw a coordinate system over the picture (you may need to enlarge the photo; a free photo editor http://www.gimp.org/downloads/).
- **a.** Mark the x and y axes.
- **b.** Mark the scale.

3. Find three points on the graph that can be used to find the equation of the parabola in the form $f(x) = a\left(x-h\right)^2 + k$.

4. Use two other points on the graph to check your equation (it only needs to be an approximation).

5. Identify the following:
- **a.** The vertex.
- **b.** All intercepts.
- **c.** The maximum and minimum.
- **d.** The domain and the range.

6. Write the equation in the form $f(x) = ax^2 + bx + c$. Show all your work.

7. Write an essay.
- **a.** Describe how you completed each of the above steps.
- **b.** Show all your work from the above steps.
- **c.** Include the picture with the coordinate system.